

# Time Series Regression

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- Data where the cases represent time: data collected every day, month, year, etc.
- Time series are important for both **explaining** how variables change over time and **forecasting** the future
- Examples of time series data:
  - Google's closing daily stock price every day in 2020
  - Inventory levels of each item at a retail store at the end of every week in 2020
  - Number of new COVID cases in the US each day since the start of the pandemic
  - Apple's quarterly revenue since 2009



Some notation:

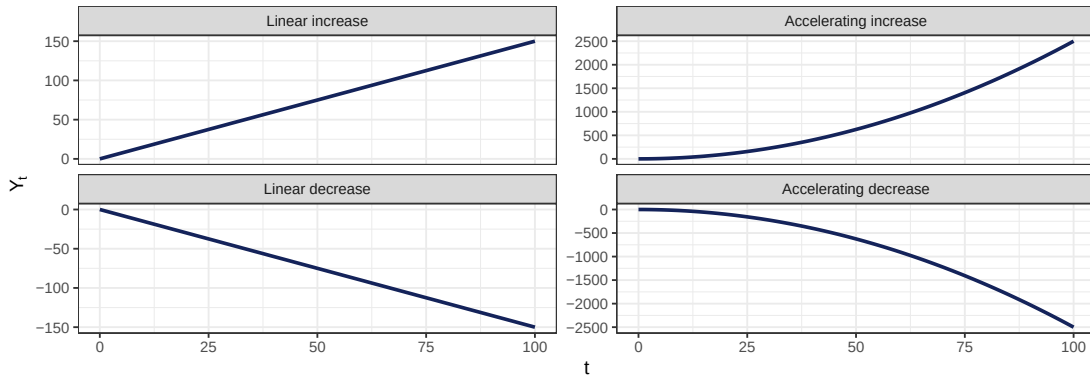
$t$  = time

$Y_t$  = the value of the variable we are interested in, at time  $t$

$Y_t$  may be composed of one or more components:

- Trend
- Seasonal
- Cyclical
- Random

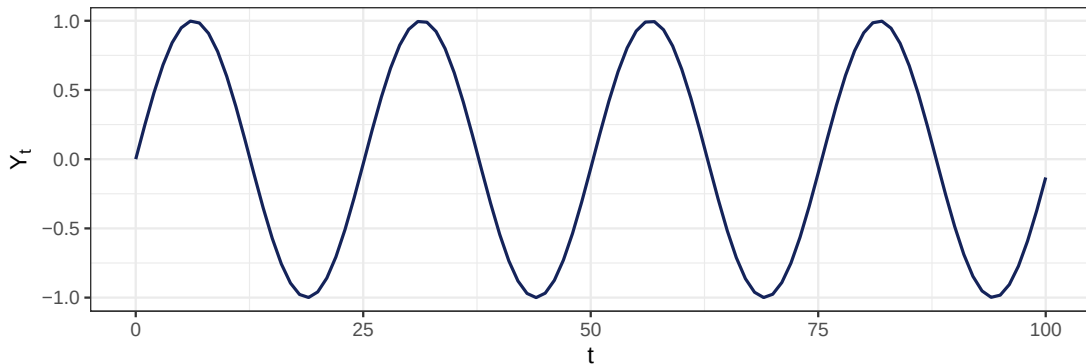
A **trend** is persistent upwards or downwards movement in the data.





- The trend doesn't need to be linear!
- Example: Moore's Law (accelerating increase of transistor count)
- Example: US population
- A time series with no trend is stationary

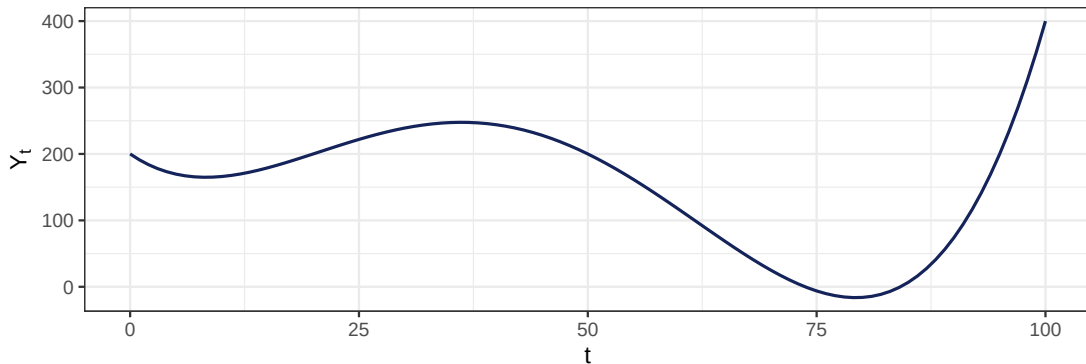
**Seasonal** fluctuation occurs when predictable up or down movements occur *over a regular interval*.





- The ups and downs must occur over a regular interval (e.g., every month, or every year)
- Example: Highway traffic volume is highest during rush hour every day
- Example: Supermarket sales may be highest every month right after common paydays like the 15th and 30th

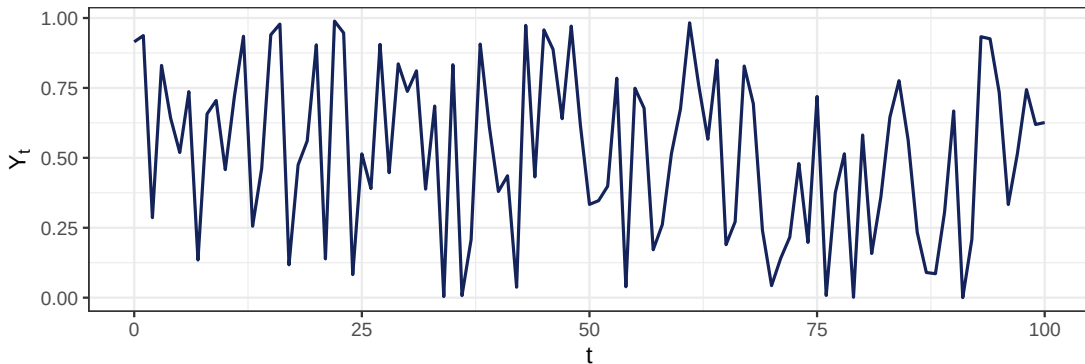
**Cyclic** fluctuations occur at unpredictable intervals, e.g. due to changing business or economic conditions.





- In contrast to seasonal fluctuations, cyclic fluctuations do not occur at regular, predictable intervals
- It may be possible to predict cyclic components based on some other (non-time) variable
- Example: Restaurant sales dropped dramatically in 2020 due to COVID, as people ate out less
- Example: Sales of bell bottoms rose in the 60s and 70s, declined by the 80s, and then had a resurgence in the 90s

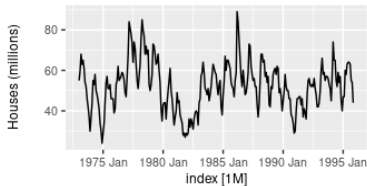
Any real time series will always have **random noise** as well, which can't be predicted or forecast.



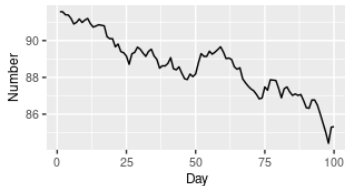
# Which component(s) do you see?



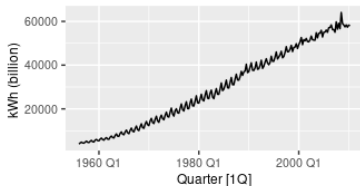
Sales of new one-family houses, USA



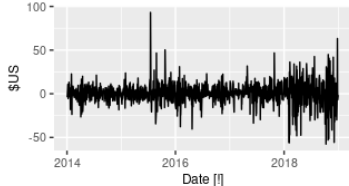
US treasury bill contracts



Australian quarterly electricity production



Daily changes in Google closing stock price





Real time series will usually include a combination of these four components. We will model the time series  $Y_t$  either additively:

$$Y_t = \text{trend} + \text{seasonal} + \text{random} = T_t + S_t + E_t$$

Or multiplicatively:

$$Y_t = \text{trend} \cdot \text{seasonal} \cdot \text{random} = T_t \cdot S_t \cdot E_t$$

( $E_t$  consists of both the cyclic and error components, as both are unpredictable.)

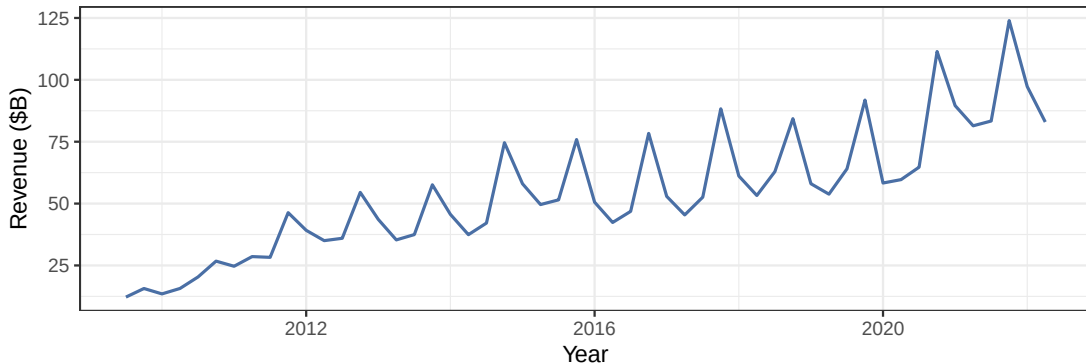


- All components  $(T_t, S_t, E_t)$  are in the units of  $Y$
- Most appropriate when seasonal fluctuations are consistent (do not increase or decrease over time)
- The trend component  $T_t$  is a function of  $t$  (e.g., linear or quadratic)
- The seasonal component  $S_t$  is a set of dummy variables representing “seasons”
- So we can estimate additive models using regular old regression!



1. Run a regression predicting  $Y$  as a function of:
  - $t$ ,  $t^2$ ,  $\log(t)$  etc (the **trend component**  $T_t$ )
  - Dummy variables for the seasons (the **seasonal component**  $S_t$ )
2. To make a prediction for  $Y$ , plug into the model!
3. The residuals of this model correspond to the **error component**  $E_t$

# Apple quarterly revenue



What components do you see here?

```
head(apple)
```

|   | Period | Time    | Year | Quarter | Revenue |
|---|--------|---------|------|---------|---------|
| 1 | 1      | 2009.50 | 2009 | Q3      | 12.21   |
| 2 | 2      | 2009.75 | 2009 | Q4      | 15.68   |
| 3 | 3      | 2010.00 | 2010 | Q1      | 13.50   |
| 4 | 4      | 2010.25 | 2010 | Q2      | 15.70   |
| 5 | 5      | 2010.50 | 2010 | Q3      | 20.34   |
| 6 | 6      | 2010.75 | 2010 | Q4      | 26.74   |

(Note that the fractional part of `Time` indicates the quarter.)

```
linear.trend <- lm(Revenue ~ Period, data=apple)
summary(linear.trend)
```

Call:

```
lm(formula = Revenue ~ Period, data = apple)
```

Residuals:

|  | Min    | 1Q    | Median | 3Q   | Max   |
|--|--------|-------|--------|------|-------|
|  | -19.88 | -8.13 | -3.24  | 6.74 | 36.20 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )    |
|-------------|----------|------------|---------|-------------|
| (Intercept) | 17.452   | 3.758      | 4.64    | 2.5e-05 *** |
| Period      | 1.406    | 0.123      | 11.39   | 1.7e-15 *** |

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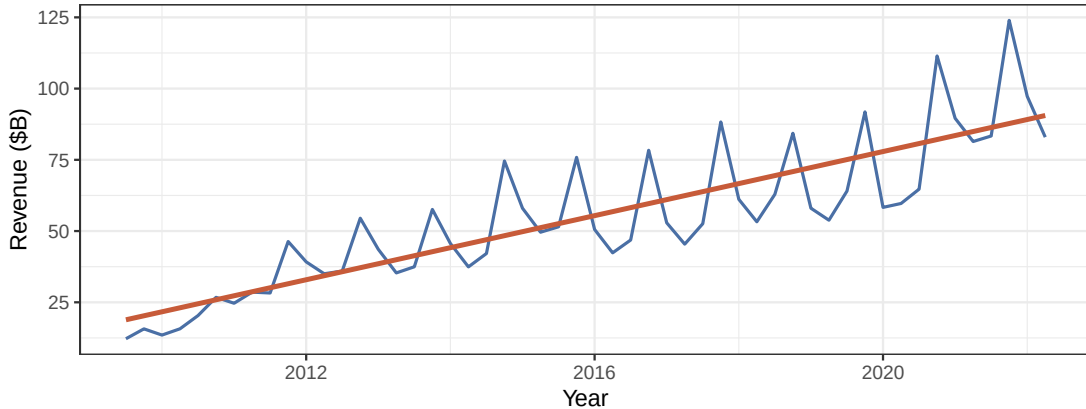
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.4 on 50 degrees of freedom

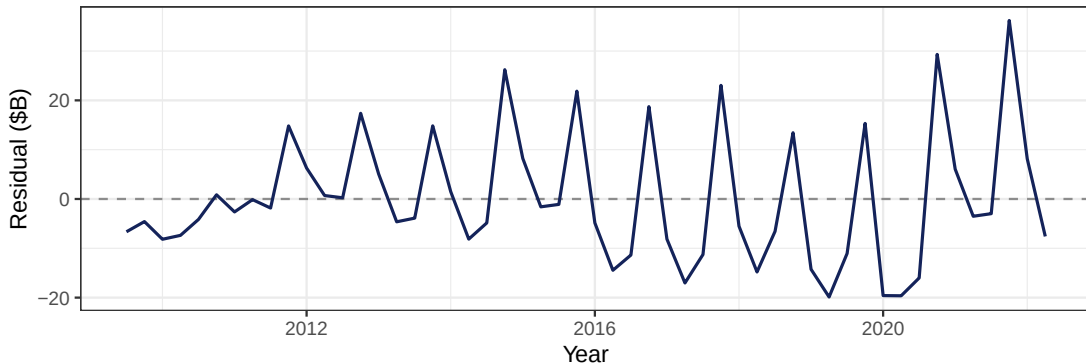
Multiple R-squared: 0.722, Adjusted R-squared: 0.716

F-statistic: 130 on 1 and 50 DF, p-value: 1.67e-15

# Fitting a linear trend component



The residuals show the “detrended” data — but there’s still some trend left, and clear seasonality we are ignoring:



Let's add dummies for the quarter (leaving one out, like any categorical variable):

```
linear.trend.seasonality <- lm(Revenue ~ Period + Quarter, data=apple)
summary(linear.trend.seasonality)
```

Call:

```
lm(formula = Revenue ~ Period + Quarter, data = apple)
```

Residuals:

| Min    | 1Q    | Median | 3Q   | Max   |
|--------|-------|--------|------|-------|
| -21.70 | -5.13 | 1.09   | 5.30 | 18.33 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )   |     |
|-------------|----------|------------|---------|------------|-----|
| (Intercept) | 14.8822  | 3.1273     | 4.76    | 0.00001892 | *** |
| Period      | 1.4215   | 0.0776     | 18.32   | < 2e-16    | *** |
| QuarterQ2   | -6.9492  | 3.2850     | -2.12   | 0.04       | *   |
| QuarterQ3   | -4.0878  | 3.2878     | -1.24   | 0.22       |     |
| QuarterQ4   | 19.6569  | 3.2850     | 5.98    | 0.00000028 | *** |

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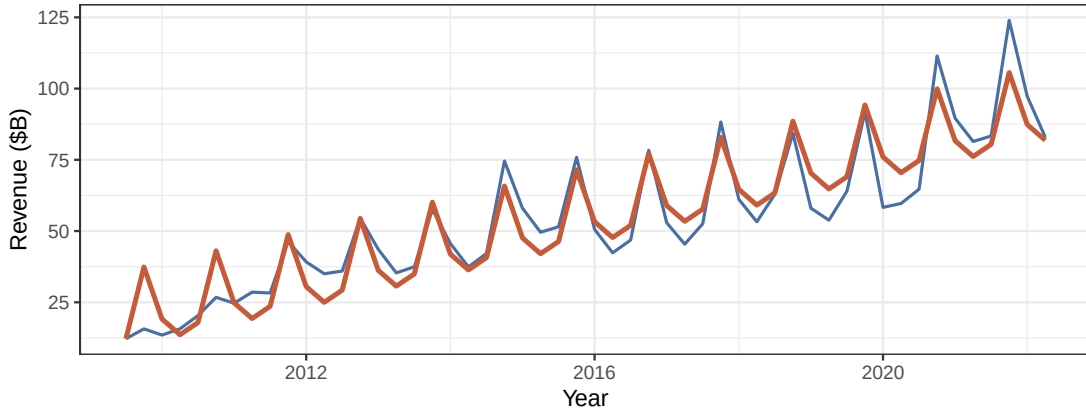
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.37 on 47 degrees of freedom

Multiple R-squared: 0.897, Adjusted R-squared: 0.889

F-statistic: 103 on 4 and 47 DF, p-value: <2e-16

# Linear trend with seasonality



|             | Estimate | Std. Error | t value | Pr(> t )   |     |
|-------------|----------|------------|---------|------------|-----|
| (Intercept) | 14.8822  | 3.1273     | 4.76    | 0.00001892 | *** |
| Period      | 1.4215   | 0.0776     | 18.32   | < 2e-16    | *** |
| QuarterQ2   | -6.9492  | 3.2850     | -2.12   | 0.04       | *   |
| QuarterQ3   | -4.0878  | 3.2878     | -1.24   | 0.22       |     |
| QuarterQ4   | 19.6569  | 3.2850     | 5.98    | 0.00000028 | *** |

- Q2's are expected to be \$6.9B worse than Q1's
- Q3's are expected to be \$4.1B worse than Q1's
- Q4's are expected to be \$19.7B better than Q1's

# What does our model predict for the future?



What does the final model predict for Apple in 2028 Q1?



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2028 Q1 is `Period` = 75, and Q1 is the reference quarter — so no seasonal term is added:

$$\begin{aligned}\widehat{\text{Revenue}} &= 14.88 + 1.42 \times 75 \\ &= 121.5 \text{ billion dollars}\end{aligned}$$



What does the final model predict for Apple in 2028 Q1?

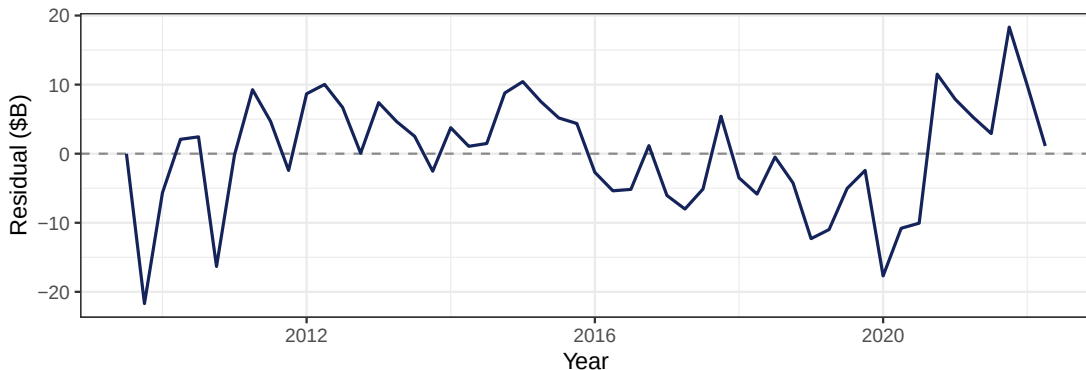
2028 Q1 is  $\text{Period} = 75$ , and Q1 is the reference quarter — so no seasonal term is added:

$$\begin{aligned}\widehat{\text{Revenue}} &= 14.88 + 1.42 \times 75 \\ &= 121.5 \text{ billion dollars}\end{aligned}$$

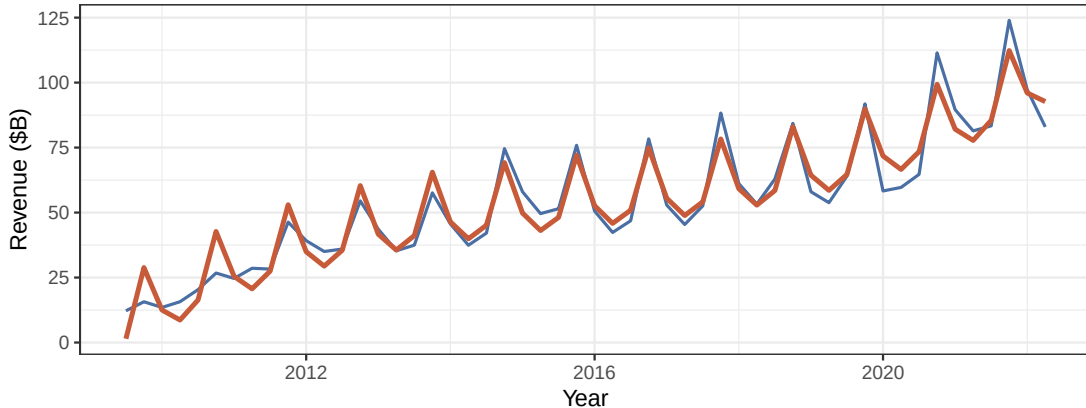
Should we trust it? That is 23 quarters past the end of the data, and the trend here is a straight line we chose.



The residuals now show the “detrended and deseasonalized” data — but there’s still some trend left:



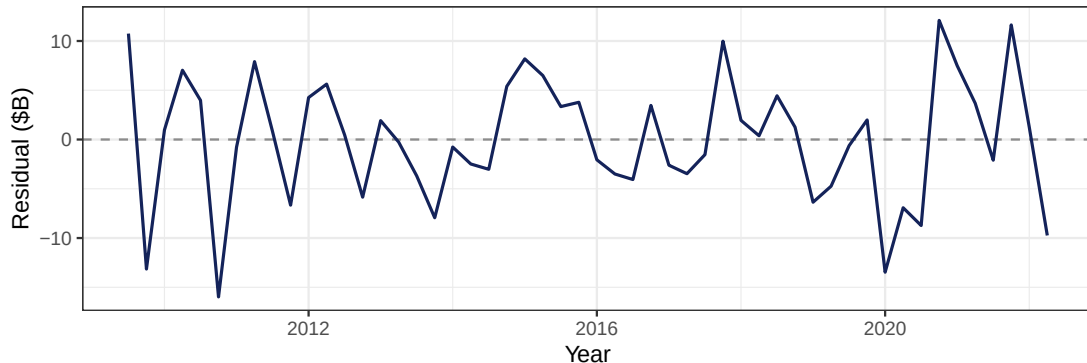
## Improving the model by fitting a cubic trend



## Detrended and deseasonalized data (cubic model)



Much better! What's left looks random:





- Only the trend component  $T_t$  is in the units of  $Y$
- The seasonal and error components are multipliers (e.g., applying a percentage rather than absolute change to the reference season)
- We can estimate this using regression by taking logs of both sides of the equation  $Y_t = T_t \cdot S_t \cdot E_t$ :

$$\log(Y_t) = \log(T_t) + \log(S_t) + \log(E_t)$$

- A multiplicative model is appropriate when the magnitude of the seasonality increases (or decreases) over time

```
log.trend.seasonality <- lm(log(Revenue) ~ log(Period) + Quarter, data=apple)
summary(log.trend.seasonality)
```

Call:

```
lm(formula = log(Revenue) ~ log(Period) + Quarter, data = apple)
```

Residuals:

|  | Min     | 1Q      | Median  | 3Q     | Max    |
|--|---------|---------|---------|--------|--------|
|  | -0.2083 | -0.0810 | -0.0236 | 0.0954 | 0.3540 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )       |
|-------------|----------|------------|---------|----------------|
| (Intercept) | 2.1936   | 0.0719     | 30.53   | < 2e-16 ***    |
| log(Period) | 0.5493   | 0.0204     | 26.95   | < 2e-16 ***    |
| QuarterQ2   | -0.1248  | 0.0507     | -2.46   | 0.017 *        |
| QuarterQ3   | -0.0453  | 0.0508     | -0.89   | 0.377          |
| QuarterQ4   | 0.3166   | 0.0507     | 6.25    | 0.00000011 *** |

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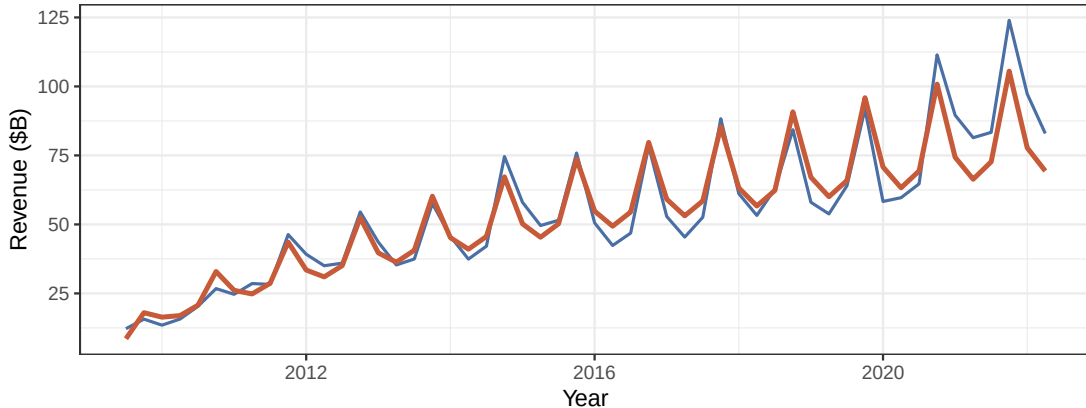
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.129 on 47 degrees of freedom

Multiple R-squared: 0.945, Adjusted R-squared: 0.94

F-statistic: 202 on 4 and 47 DF, p-value: <2e-16

# Multiplicative model captures increasing variation





Our model of the trend component is:

$$\log(\widehat{\text{Rev}}) = 2.19 + 0.55 \log(t) - 0.12(\text{Q2}) - 0.05(\text{Q3}) + 0.32(\text{Q4})$$

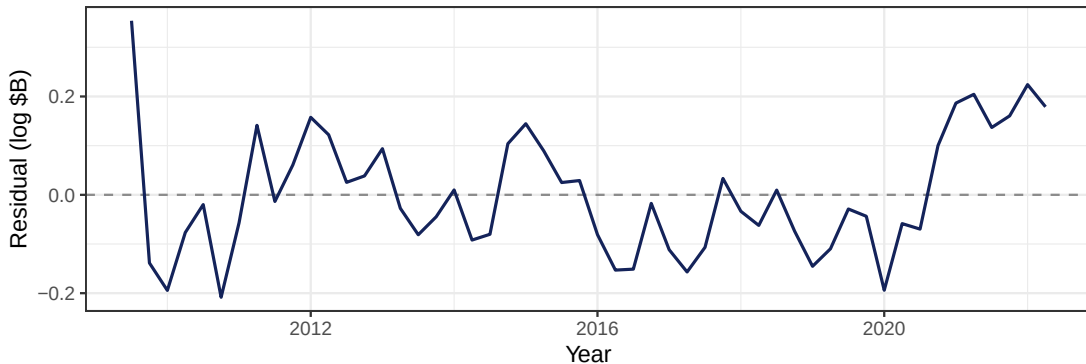


|             | Estimate | Std. Error | t value | Pr(> t )   |     |
|-------------|----------|------------|---------|------------|-----|
| (Intercept) | 2.1936   | 0.0719     | 30.53   | < 2e-16    | *** |
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| QuarterQ3   | -0.0453  | 0.0508     | -0.89   | 0.377      |     |
| QuarterQ4   | 0.3166   | 0.0507     | 6.25    | 0.00000011 | *** |

Because this is a model where we are predicting  $\log(Y)$ :

- Q2's are expected to be 12% below Q1's
- Q3's are expected to be 5% below Q1's
- Q4's are expected to be 32% above Q1's

Good! What's left looks (pretty) random:

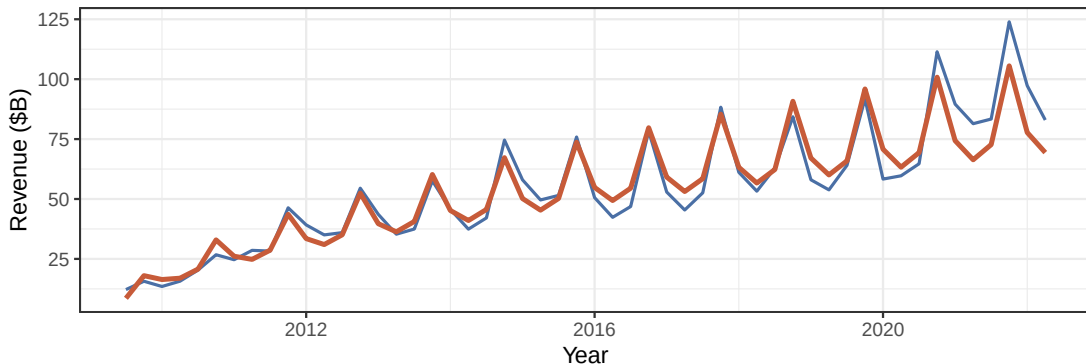


Sometimes forecasting using a trend-based model is dangerous—things change!

# Have things changed?



Apple seems to be consistently outperforming the model in the last couple of years.





Previously, we created **decomposition** models that predict  $Y_t$  as a function of time:

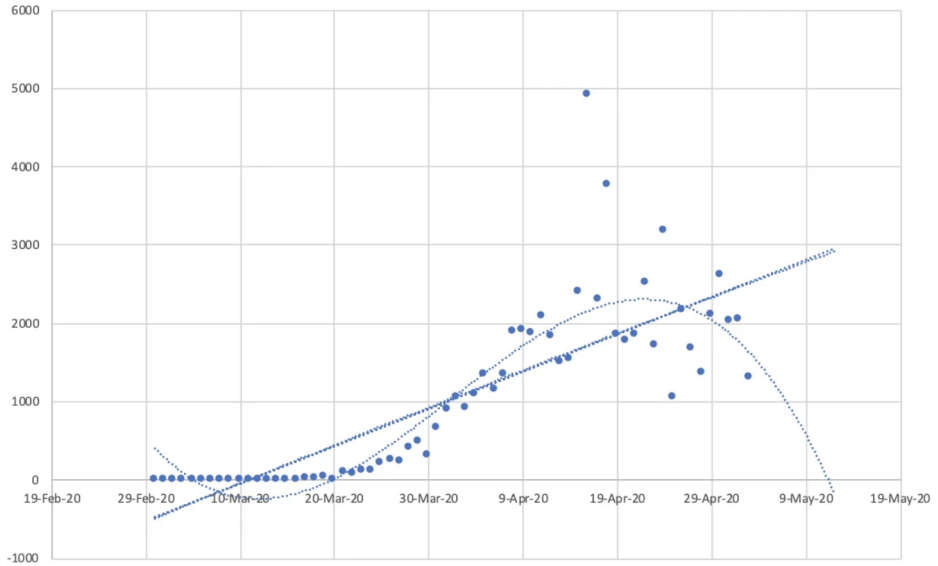
$$Y_t = \text{trend} + \text{seasonal} + \text{error} = T_t + S_t + E_t$$

$$Y_t = \text{trend} \times \text{seasonal} \times \text{error} = T_t \times S_t \times E_t$$

But sometimes forecasting using a trend-based model is dangerous or unworkable!

We might also be leaving some predictive ability on the table if  $E_t$  includes a cyclical component.

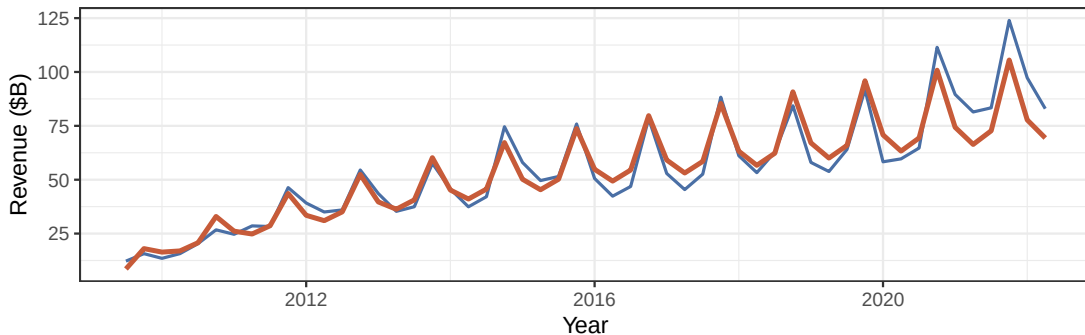
Covid Death Scatterplot With Linear and Cubic Trendlines



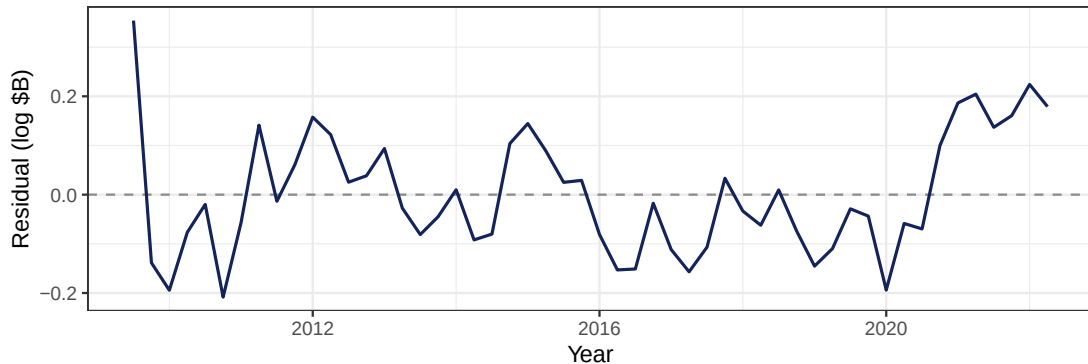
## Recall: Multiplicative model for Apple's revenue



```
lm(log(Revenue) ~ log(Period) + Quarter, data=apple)
```

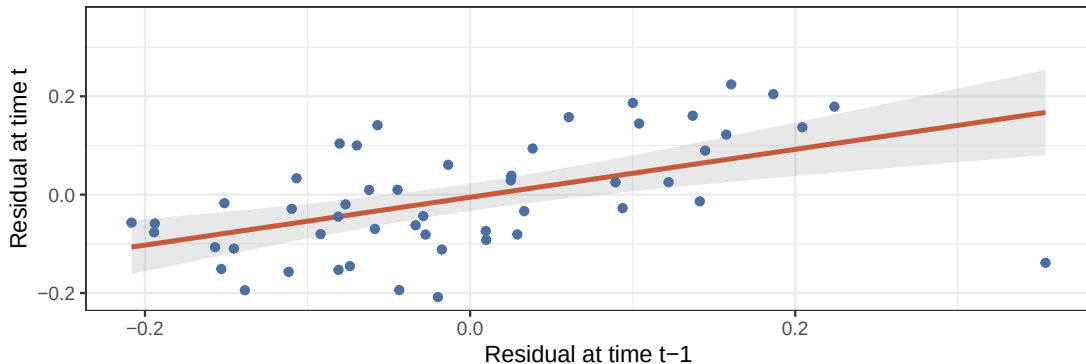


## Recall: Multiplicative model for Apple's revenue



We thought our residuals looked pretty good, but...

## Past prediction errors predict future errors

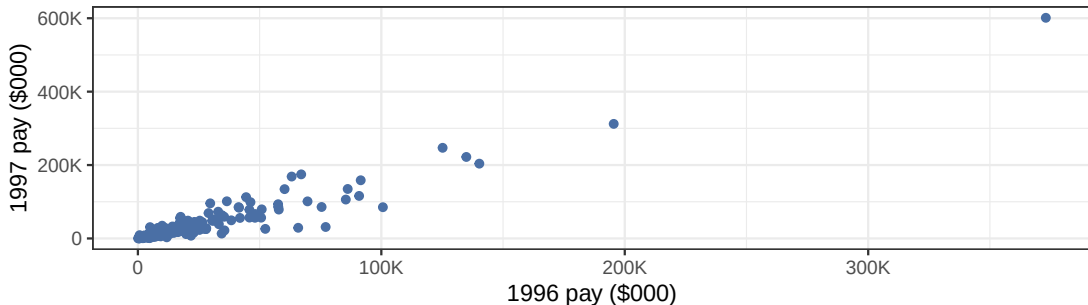


How can we use this to improve our model?

What factors do you think would be the best predictors of how much a company pays their CEO in a given year?

What factors do you think would be the best predictors of how much a company pays their CEO in a given year?

If we want to predict CEO pay in 1997, knowing the CEO's pay in 1996 seems to help a lot:



A model like this incorporates a lot of other variables that might not be available in our data set!



Key idea: Predict  $Y_t$  as a function of  $Y_{t-1}$ :

$$\hat{Y}_t = \hat{\beta}_0 + \hat{\beta}_1 Y_{t-1}$$

- $Y_{t-1}$  is called the “1st lag” of  $Y$  (e.g., if  $t = 1997$ , then  $Y_t = \text{pay1997}$  and  $Y_{t-1} = \text{pay1996}$ )



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- This is called **autoregressive** because it predicts the values of a time series based on previous values

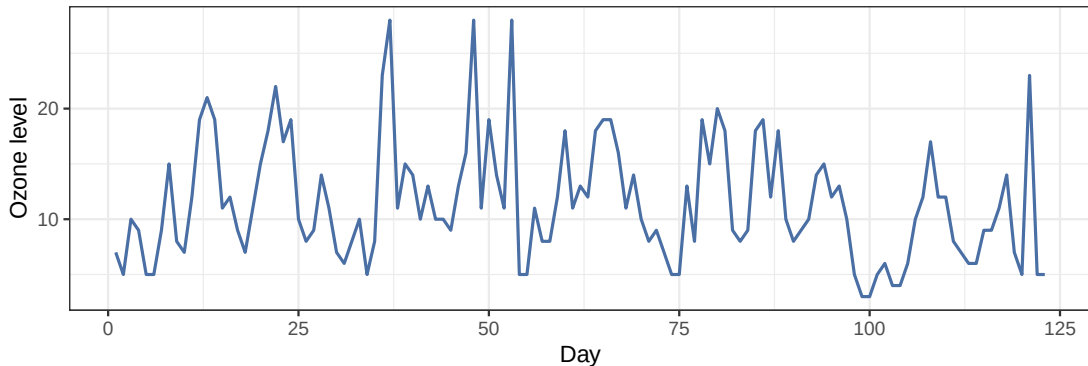


Key idea: Predict  $Y_t$  as a function of  $Y_{t-1}$ :

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- $Y_{t-1}$  is called the “1st lag” of  $Y$  (e.g., if  $t = 1997$ , then  $Y_t = \text{pay1997}$  and  $Y_{t-1} = \text{pay1996}$ )
- This is called **autoregressive** because it predicts the values of a time series based on previous values
- This is an AR(1) model (an AR( $k$ ) model would predict 1997 pay from the previous  $k$  years)

## AR(1) example: Daily ozone in Houston



Decomposing  $Y_t$  into trend and seasonal components probably isn't going to work—no clear trend or seasonality!

```
summary(lm(ozone ~ day, data=ozone))
```

Call:

```
lm(formula = ozone ~ day, data = ozone)
```

Residuals:

| Min   | 1Q    | Median | 3Q   | Max   |
|-------|-------|--------|------|-------|
| -7.97 | -4.04 | -1.09  | 2.79 | 16.27 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | 13.0152  | 0.9772     | 13.32   | <2e-16 *** |
| day         | -0.0242  | 0.0137     | -1.77   | 0.079 .    |

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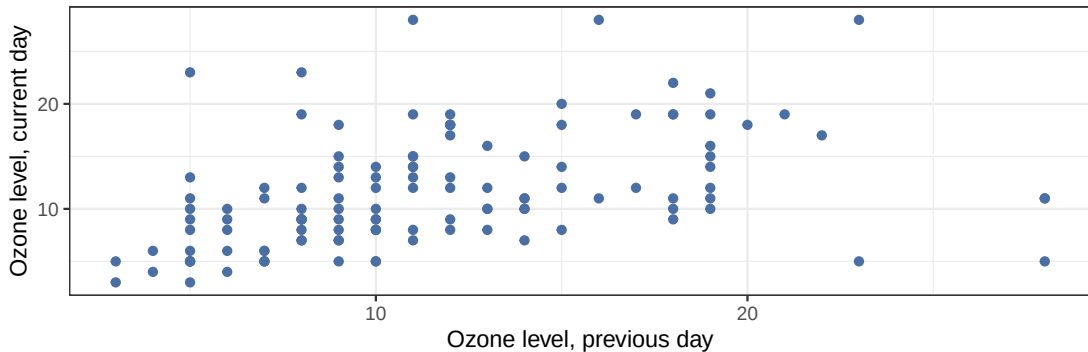
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.39 on 121 degrees of freedom

Multiple R-squared: 0.0253, Adjusted R-squared: 0.0173

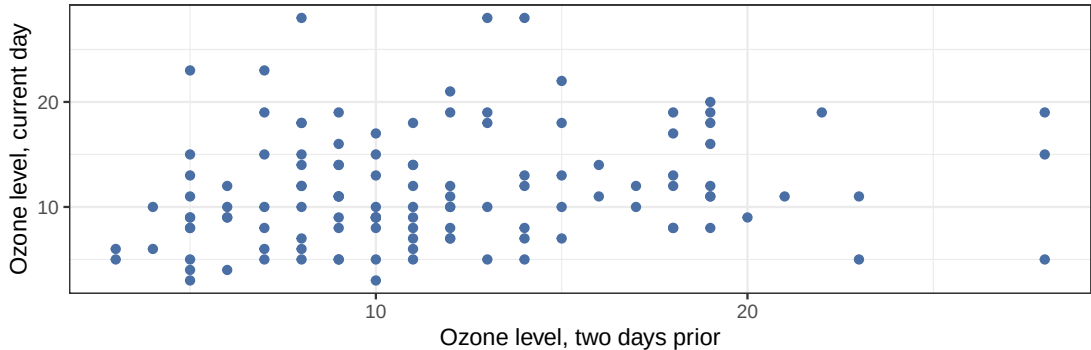
F-statistic: 3.14 on 1 and 121 DF, p-value: 0.0788

Can we use yesterday's ozone level to predict today's?



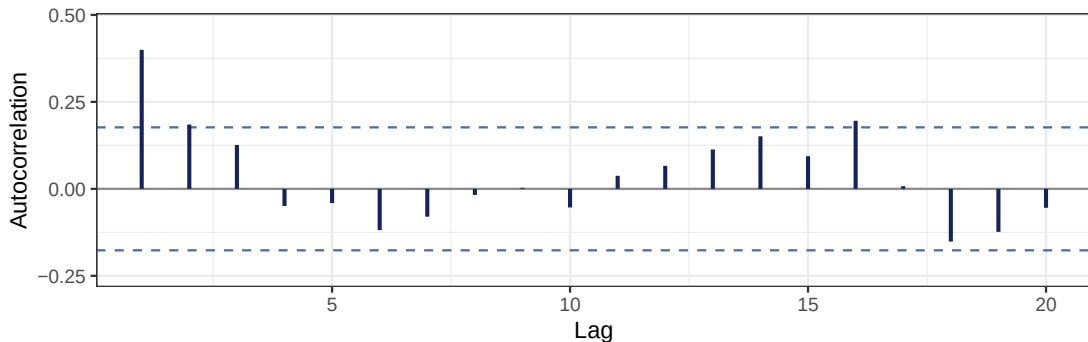
$$r = 0.403$$

How about ozone from two days ago?



$r = 0.189$  (But can you think of any reason this correlation might be misleading?)

**Autocorrelation** is the correlation of  $Y_t$  with its lags ( $Y_{t-1}$ ,  $Y_{t-2}$ , etc). The autocorrelation function — `acf(ozone$ozone)` — displays these:



Autocorrelations outside of the dashed blue lines are statistically significant.

```

ozone <- ozone %>% mutate(lag1=lag(ozone), lag2=lag(ozone, 2))
ozone.model <- lm(ozone ~ lag1, data=ozone)
summary(ozone.model)

```

```

Call:
lm(formula = ozone ~ lag1, data = ozone)

```

Residuals:

| Min    | 1Q    | Median | 3Q   | Max   |
|--------|-------|--------|------|-------|
| -13.19 | -3.46 | -1.11  | 2.68 | 16.68 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )         |
|-------------|----------|------------|---------|------------------|
| (Intercept) | 6.8745   | 1.0698     | 6.43    | 0.0000000028 *** |
| lag1        | 0.4042   | 0.0838     | 4.82    | 0.0000041974 *** |

```

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Residual standard error: 5 on 120 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.162, Adjusted R-squared: 0.155

F-statistic: 23.3 on 1 and 120 DF, p-value: 0.0000042

```
ozone.model2 <- lm(ozone ~ lag1 + lag2, data=ozone)
summary(ozone.model2)
```

Call:

```
lm(formula = ozone ~ lag1 + lag2, data = ozone)
```

Residuals:

|  | Min    | 1Q    | Median | 3Q   | Max   |
|--|--------|-------|--------|------|-------|
|  | -12.87 | -3.35 | -1.12  | 2.73 | 16.54 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )       |
|-------------|----------|------------|---------|----------------|
| (Intercept) | 6.7241   | 1.2604     | 5.33    | 0.00000047 *** |
| lag1        | 0.3835   | 0.0919     | 4.17    | 0.00005763 *** |
| lag2        | 0.0370   | 0.0922     | 0.40    | 0.69           |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.02 on 118 degrees of freedom  
(2 observations deleted due to missingness)

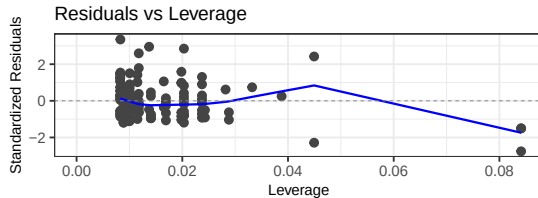
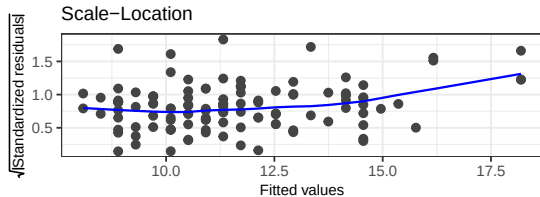
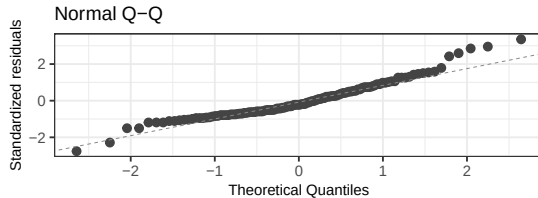
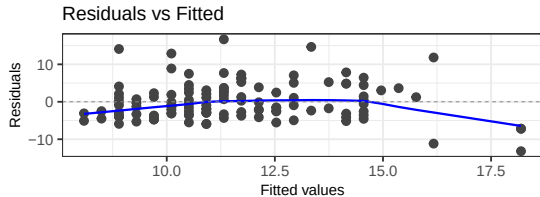
Multiple R-squared: 0.16, Adjusted R-squared: 0.145

F-statistic: 11.2 on 2 and 118 DF, p-value: 0.000035

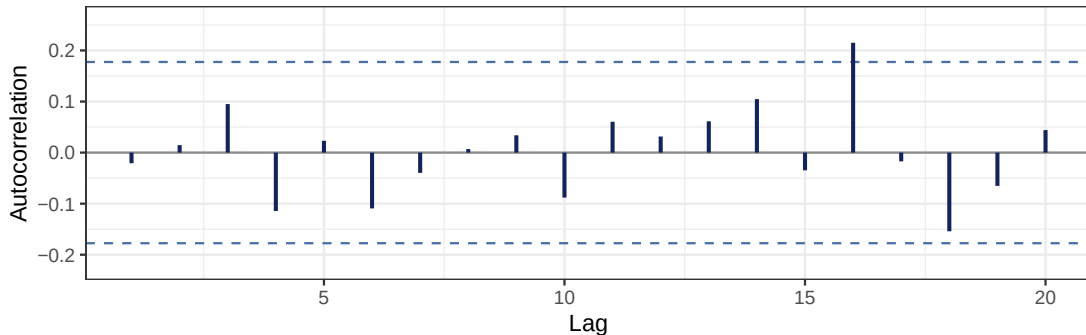


- Linearity, Homoscedasticity, Normality: Check using residual plot (linearity + homoscedasticity), Q-Q plot (normality), scale/location (homoscedasticity) like any other regression model
- Independence: Since this is a time series, we can actually check this by looking at the autocorrelation of the *residuals* (we want *no* significant autocorrelation)

# Checking assumptions of an AR(1) model



The ACF of the *residuals* — `acf(residuals(ozone.model))`:



We expect 5% of autocorrelations to be significant just by chance at  $\alpha = 0.05$ , so this ACF suggests we are OK on independence!

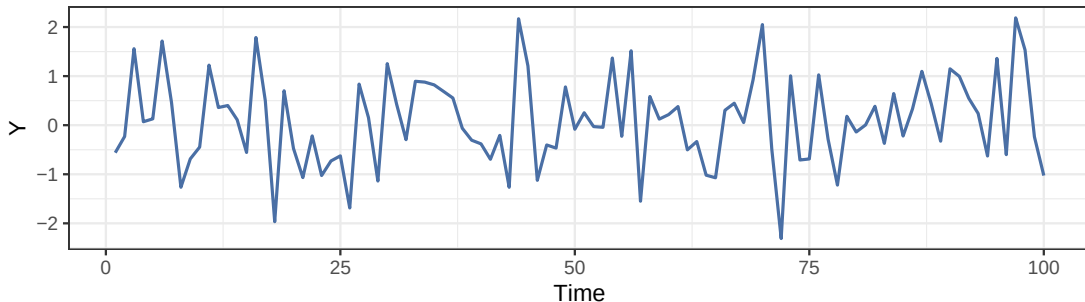


$$Y_t = \beta_0 + \beta_1 Y_{t-1} + \epsilon_t$$

- If  $\beta_0 = 0$  and  $\beta_1 = 0$ , then this is **white noise**
- If  $\beta_0 \neq 0$  and  $\beta_1 = 0$ , then this is a **random sample**
- If  $\beta_0 = 0$  and  $\beta_1 = 1$ , then this is a **random walk**
- If  $\beta_0 \neq 0$  and  $\beta_1 = 1$ , then this is a **random walk with drift**
- If  $\beta_1 < 1$ , then  $Y_t$  will oscillate around the mean

A white noise time series is one where there is no pattern at all—each value is totally unpredictable:

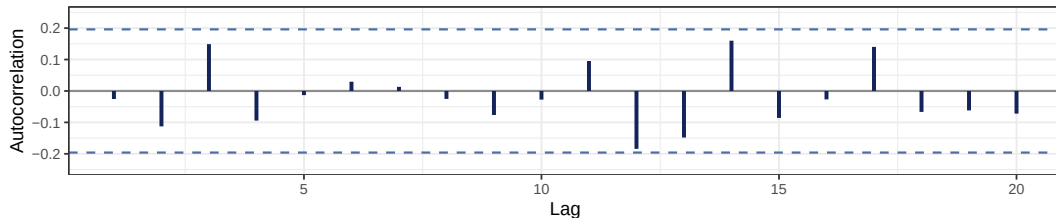
$$Y_t = \epsilon_t$$



# Could my time series be white noise? If:



1. The ACF shows no significant autocorrelation:



2. And we fit an AR(1) model and find that 0 falls within the 95% CIs for both  $\beta_0$  and  $\beta_1$ :

|             | 2.5 %   | 97.5 % |
|-------------|---------|--------|
| (Intercept) | -0.0849 | 0.284  |
| lag.y       | -0.2286 | 0.176  |

Then there is no statistically significant evidence that the series isn't white noise.

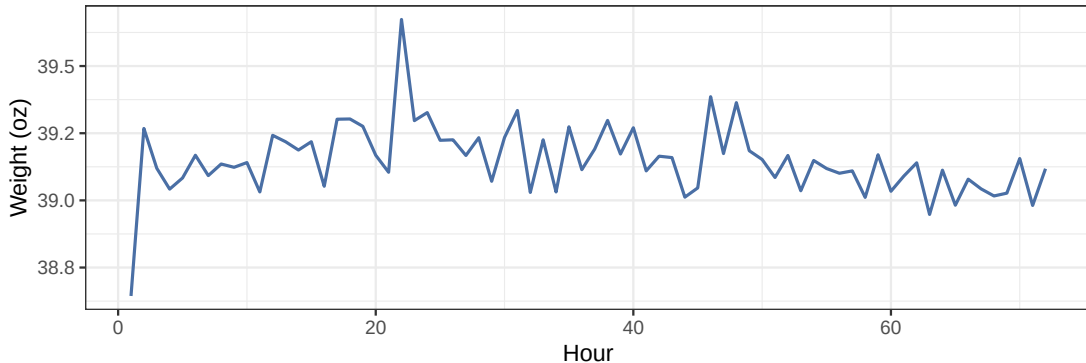


A random sample time series is one where each value is just independently sampled from the same distribution:

$$Y_t = \beta_0 + \epsilon_t$$

- There is no  $Y_{t-1}$  term: like with white noise, it doesn't matter what the previous value was
- $\beta_0$  represents the mean of the distribution that the values are being sampled from

In the Folgers factory, 39-oz coffee cans are filled up with slightly different amounts of coffee each hour (just due to random variation with the machine):



1. The ACF shows no significant autocorrelation
2. We fit an AR(1) model and find that 0 falls within the 95% CI for  $\beta_1$  but *not* for  $\beta_0$ :

```
      2.5 % 97.5 %  
(Intercept) 27.482 44.112  
lag.wt      -0.127  0.298
```

Then:

- There is **statistically significant evidence** that the series **isn't** white noise
- There is **no statistically significant evidence** that the series **isn't** a random sample



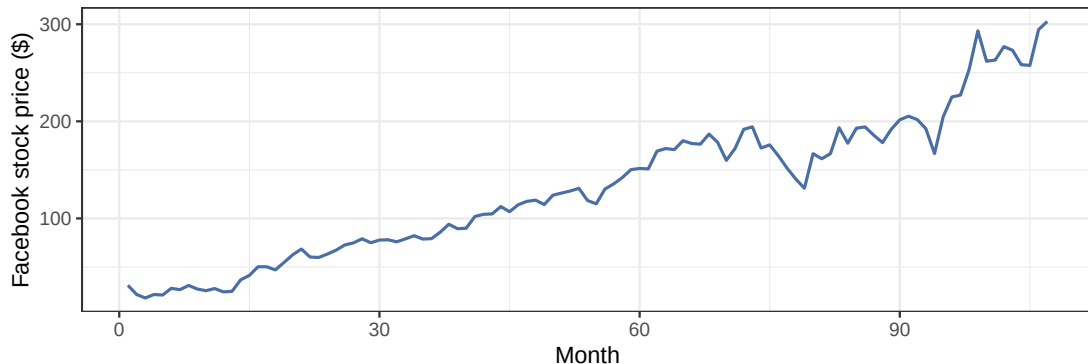
A random walk time series is one where each value is predicted to be the previous value, plus some drift (maybe):

$$Y_t = \beta_0 + Y_{t-1} + \epsilon_t$$

- $\beta_0$  is the “drift” (could be 0) and is in some sense the overall trend
- A time series is a random walk if the *changes* are a random sample time series:

$$Y_t - Y_{t-1} = \beta_0 + \epsilon_t$$

Random walks are often a good choice for stock prices, like the Facebook stock price over a 107-month period from June 2012 to last year:



We fit an AR(1) model and find that **1** falls within the 95% CI for  $\beta_1$  (and zero doesn't):

|             | 2.5 %  | 97.5 % |
|-------------|--------|--------|
| (Intercept) | -2.766 | 6.64   |
| lagFB       | 0.973  | 1.04   |

Then:

- There is **statistically significant evidence** that the series **isn't** white noise or a random sample
- There is **no statistically significant evidence** that the series **isn't** a random walk

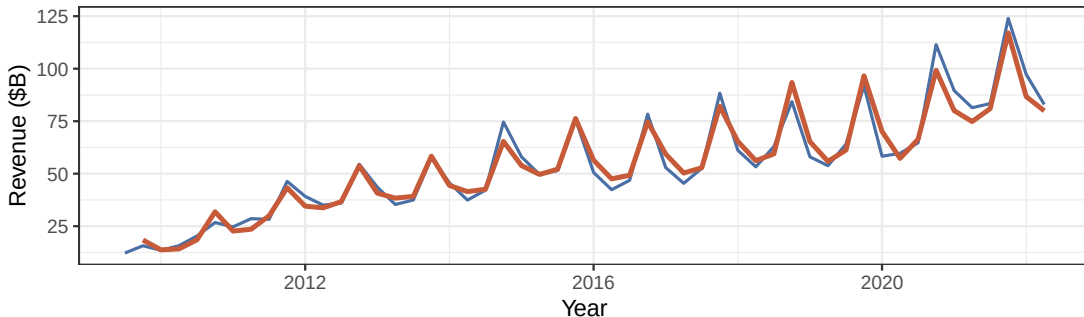
Bonus: If this series **isn't** truly a random walk, do you think it might behave similarly to a random walk anyway? Why?

| Type          | Model                                          | Predicted $Y_t$                         |
|---------------|------------------------------------------------|-----------------------------------------|
| White noise   | $Y_t = \epsilon_t$                             | 0                                       |
| Random sample | $Y_t = \beta_0 + \epsilon_t$                   | $\hat{\beta}_0$ (or average $Y$ )       |
| Random walk   | $Y_t = \beta_0 + Y_{t-1} + \epsilon_t$         | $\hat{\beta}_0 + Y_{t-1}$               |
| General AR(1) | $Y_t = \beta_0 + \beta_1 Y_{t-1} + \epsilon_t$ | $\hat{\beta}_0 + \hat{\beta}_1 Y_{t-1}$ |

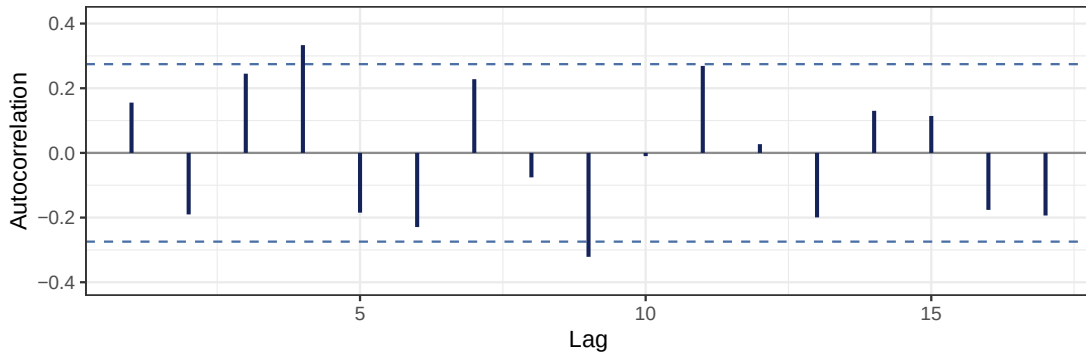
# Combining decomposition and AR models



```
apple <- apple %>% mutate(lag1 = lag(Revenue))  
apple_model <- lm(log(Revenue) ~ log(Period) + Quarter  
                  + log(lag1), data=apple)
```



The ACF of `residuals(apple_model)` is now clean:





- We started with the observation that the residuals (prediction errors) were autocorrelated. So why not use lagged *residuals* as predictors?
- We need a model to compute residuals, so the model can't depend on the residuals!
- $Y_{t-1}$  (in this case  $\log(Y_{t-1})$ ) contains all the relevant predictive information to eliminate lag-1 autocorrelation in the residuals



- Start with obvious seasonal/trend components and choose an additive or multiplicative model. (Plot your data!)
- Examine the usual diagnostic plots, and plot your residuals as a function of time. Do you need a (different) nonlinear time trend? A transformation of  $Y$ ?
- Check your residuals for autocorrelation. If it's present, add appropriate lag terms to your model.



- One final model is of interest: the **market model** for a stock estimates its volatility/risk relative to the market as a whole
- Important case of regressing **one time series on another**
- Predict the stock's percentage change (return) as a function of the percentage change of a market index (e.g., DJIA, S&P 500)
- The slope coefficient is the “beta” of the stock:
  - $\beta > 1$  means the stock is riskier/more volatile than the market as a whole
  - $\beta = 1$  means the stock is as risky/as volatile as the market as a whole
  - $\beta < 1$  means the stock is less risky/less volatile than the market as a whole



First let's calculate the percentage returns:

```
dell <- dell %>%  
  mutate(dell.return=(dell-lag(dell))/lag(dell),  
         SP500.return=(SP500-lag(SP500))/lag(SP500))
```

Now let's predict Dell's returns from the S&P 500:

```
dell.model <- lm(dell.return ~ SP500.return, data=dell)
summary(dell.model)
```

Call:

```
lm(formula = dell.return ~ SP500.return, data = dell)
```

Residuals:

| Min      | 1Q       | Median   | 3Q      | Max     |
|----------|----------|----------|---------|---------|
| -0.05157 | -0.00663 | -0.00134 | 0.00912 | 0.03577 |

Coefficients:

|              | Estimate | Std. Error | t value | Pr(> t )       |
|--------------|----------|------------|---------|----------------|
| (Intercept)  | -0.00217 | 0.00183    | -1.18   | 0.24           |
| SP500.return | 1.09719  | 0.18946    | 5.79    | 0.00000026 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.0144 on 61 degrees of freedom  
(1 observation deleted due to missingness)

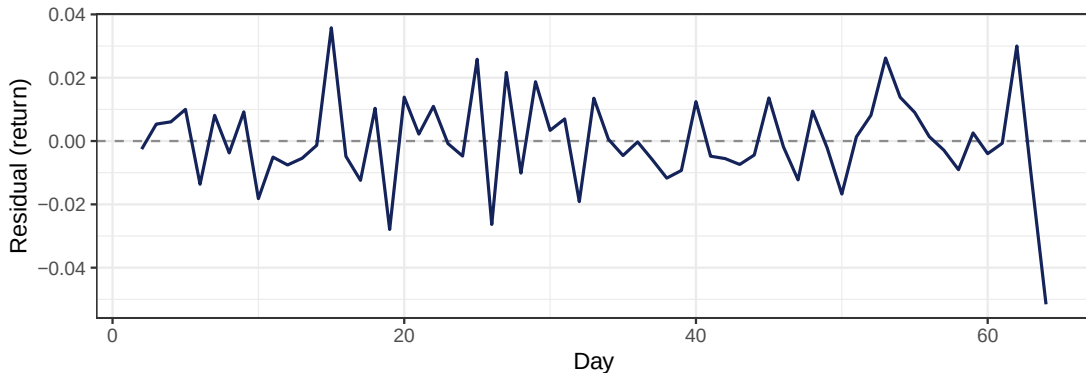
Multiple R-squared: 0.355, Adjusted R-squared: 0.344

F-statistic: 33.5 on 1 and 61 DF, p-value: 0.000000262

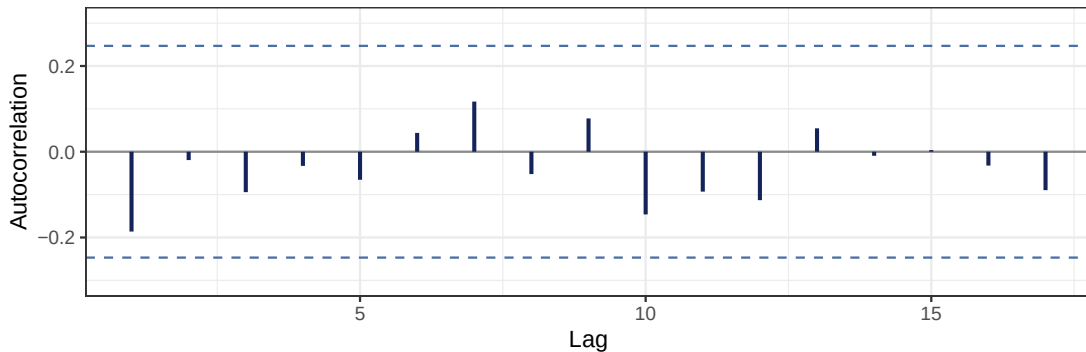
## Do the residuals depend on time?



Standard diagnostics look pretty good. How about time dependence in residuals?



The ACF of `resid(dell.model)`:



These look good!



- Dell's  $\beta$  of 1.10 indicates it's about as volatile/risky as the market as a whole
- The  $R^2$  of 0.35 indicates that about 35% of the variability in Dell's stock price can be attributed to the market's movement
- Since  $1 - R^2 = 0.65$ , about 65% of the variability in Dell's stock price is from Dell-specific (or industry-specific) factors